

## **DRAWING DESCRIPTIONS — PATENT A**

### **Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems**

#### **BRIEF DESCRIPTION OF THE DRAWINGS**

**FIG. 1** is a system architecture block diagram showing the eight-layer self-sustaining neural-motor energy harvesting loop.

**FIG. 2** is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2000 operational steps.

**FIG. 3** is a schematic diagram of the spiking neural network architecture showing the leaky integrate-and-fire neuron model with synaptic connections.

**FIG. 4** is a circuit schematic of the energy harvesting subsystem comprising piezoelectric and thermoelectric transducers with rectification and storage.

**FIG. 5** is a graph showing the activity-dependent energy dynamics curve illustrating the quadratic cost model and optimal activity operating region.

**FIG. 6 is a hardware reference design layout diagram showing component placement and interconnections for the physical embodiment.**

**FIG. 7 is a validation results summary table confirming all falsifiable claims pass the specified thresholds.**

**FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the canonical signal flow from sensor interface through neuromorphic core, recursive reflection, energy governor, cognitive modulation, neural router, and actuation, with energy harvesting feedback.**

## **DETAILED DESCRIPTION OF THE DRAWINGS**

**Reference Numeral Convention:** Each figure in this patent uses a per-figure numbering series (FIG. 1 uses 100-series, FIG. 2 uses 200-series, through FIG. 8 uses 800-series), with even-numbered increments. Where the same physical component appears in multiple figures, it receives the numeral appropriate to that figure's series.

## **FIG. 1 — System Architecture Block Diagram (8-Layer Loop)**

**This figure shows eight labeled blocks arranged in a vertical flow with arrows indicating signal direction, forming a closed loop.**

**Block layout (top to bottom, then looping back):**

**ENVIRONMENT 100 (external, at top)**

|

v

**L1: Printed Membrane 102 — labeled "ThermochromicMixin"**

|

v

**L2: Sensing Pads 104 — labeled "light\_intensity, membrane\_temp"**

|

v

**L3: Optical Transmission 106 — labeled "signal\_voltage = light / 1000 x 5.0"**

|

**v**

**L4: Neuromorphic CPU (SNN) 108 — labeled "BaseSNN (LIF neurons)"**

| Arrow entering from L8: Recursive

| Reflection 116 labeled "reflection feedback"

**v**

**L5: Servo Actuation 110 — labeled "target\_angle = clip(output x 180)"**

|

+-----> **L6: Energy Harvesting 112 — labeled "piezo + thermal"**

|

**v**

**L7: Ground Reference 114 — labeled "noise floor / baseline"**

|

v

**L8: Recursive Reflection 116 — labeled "previous\_output"**

|

**+-----> Recursive Feedback 120 (arrow loops back to**

**Neuromorphic CPU 108)**

**Three signal pathways should be distinguished by line style:**

**- Solid line: Main spine (Printed Membrane 102 through Recursive**

**Reflection 116 to Neuromorphic CPU 108)**

**- Dashed line: Thermal Cross-Link 118 (Printed Membrane 102 membrane\_temp**

**directly to Energy Harvesting 112)**

**- Dotted line: Recursive Feedback 120 (Recursive Reflection 116 to**

**Neuromorphic CPU 108)**

All blocks are rectangles with rounded corners. Environment 100 is shown as a cloud shape. The loop-back arrow (Recursive Feedback 120) from Recursive Reflection 116 to Neuromorphic CPU 108 should be prominent and labeled "THE LOOP CLOSES HERE."

**FIG. 2 — Energy Balance Comparison**

This figure shows a dual-panel line graph or single graph with two curves.

**X-axis: Operational Steps (0 to 2000)**

**Y-axis: Cumulative Energy (mWh)**

**Experimental curve 200 (dashed line, labeled "BalancedEnergyConfig (Experimental)"):**

- Starts at 50.0 mWh
- Rises rapidly to 100.0 mWh (storage capacity) within approximately 30 steps
- Plateaus at 100.0 mWh for remaining steps (homeostatic equilibrium)
- Label: "base\_consumption = 45 mW, capacity = 100 mWh, overflow → heat"

**Control curve 202 (solid line, labeled "EnergyConfig (Control)":**

- Starts at 50.0 mWh
- Decreases linearly
- Ends at approximately -4,697 mWh at step 2000
- Label: "base\_consumption = 470 mW"

**A horizontal dashed line at  $y = 0$  serves as the zero energy baseline 204,**

**labeled "Self-Sustaining Threshold."**

**A horizontal dashed line at  $y = 100$  serves as the capacity limit 206, labeled**

**"Storage Capacity (100 mWh)."**

**Annotations:**

- Arrow pointing to control curve 202: "System fails — energy depleted"
- Arrow pointing to experimental curve 200 plateau: "System achieves homeostasis at capacity"

The area between experimental curve 200 and control curve 202 should be lightly shaded to emphasize the divergence.

### **FIG. 3 — SNN Architecture (LIF Neuron Model)**

This figure shows the internal structure of the spiking neural network.

#### **Top section — Single LIF Neuron Detail:**

- Circle labeled Neuron i 308 (LIF model)
- Input arrows entering from left: I\_external input 302 and I\_reflection input 304
- Both inputs converge at summing junction 306 before entering Neuron i 308
- Inside the circle: equation written as " $V(t+1) = V(t) \times \text{leak} + I \times dt$ "
- V\_th threshold 310: horizontal dashed line labeled " $V_{\text{threshold}} = 0.5$ "
- Spike output 312 arrow exiting right: "spike (0 or 1)"
- Below Neuron i 308: "If  $V \geq \text{threshold}$ : spike,  $V = V_{\text{reset}}$ "

#### **Bottom section — Network Topology:**



- **Network neurons 314:** N circles (labeled "1" through "N") arranged in a single layer
- **All-to-all connections shown as lines between network neurons 314**
- **$W[i,j]$  weight matrix 316 notation on one representative connection**
- **Input arrow from left entering all network neurons 314: "adjusted\_input"**
- **spike\_vector output 318 arrows from all network neurons 314 going right**
- **Self-connections shown as small loops, crossed out (zeroed)**
- **Label: "N = 50 neurons (operational), N = 20 neurons (control)"**

**Inset box showing STDP Learning Rule 320:**

- **"If post fires:  $W += A+ \times \text{pre\_trace}$  (LTP)"**
- **"If pre fires:  $W -= A- \times \text{post\_trace}$  (LTD)"**
- **" $A- > A+$  (stability bias)"**

#### **FIG. 4 — Energy Harvesting Circuit**

This figure shows an electrical circuit schematic.

**Left side — Piezoelectric Harvester:**

- **Piezoelectric disc 400 (capacitor with "PIEZO" label)**
- **Connected to servo motor 416 shaft via mechanical coupling 406**  
(zigzag line)
- **Label: "27mm piezoelectric disc"**
- **Output leads to full-bridge rectifier 402 (4 diodes in bridge configuration)**
- **Full-bridge rectifier 402 output to smoothing capacitor 404**
- **Label: "friction\_factor = 18.0"**

**Center — LC Resonance Circuit:**

- **Ferrite inductor 418 in series with piezoelectric disc 400 output**
- **Label: "Laird ferrite-backed inductor"**
- **Resonance annotation: " $f_{\text{res}} = 1 / (2 \times \pi \times \sqrt{L \times C})$ "**
- **Label: "3-10x voltage amplification at resonance"**

**Right side — Thermoelectric Harvester:**

- NTC 10K 3950 thermistor 408
- Connected to thermoelectric rectifier 410
- Thermoelectric rectifier 410 output to thermoelectric smoothing capacitor 412
- Label: "thermal\_factor = 8.0"

**Bottom — Energy Storage:**

- Both harvester outputs combine at power combiner 420
- Arrow to energy storage 414 (battery/capacitor symbol)
- Label: "energy\_stored (mWh)"
- Output arrow labeled "Power to SNN (L4)"

**Power budget annotation box:**

- "Harvest = friction\_harvest + thermal\_harvest"

- " $\text{Cost} = \text{base\_consumption} + \text{activity\_cost} \times \text{spike\_count}^2$ "

- " $\text{Net} = \text{Harvest} - \text{Cost}$ "

### **FIG. 5 — Activity-Dependent Energy Dynamics**

**This figure shows a graph derived from `BalancedEnergyConfig` parameters (core/energy.py) illustrating the energy harvesting and consumption curves as functions of neural spike count.**

**X-axis: "Neural Activity (Spikes / Step)" ranging from 0 to 50**

**Y-axis: "Energy per Step (mWh)"**

**Two curves:**

**1. Harvest curve (ref 500, dashed line): Nearly linear, gently increasing**

- **Formula:  $\text{Harvest} = 0.20 + (0.09 \times \text{spikes})$  mWh/step**

- **Composed of thermal baseline (0.20 mWh at  $\Delta T = 5$  degrees C) plus**

**piezoelectric harvest proportional to spike-driven motor activity**

- **Rises from 0.20 mWh at 0 spikes to approximately 4.7 mWh at 50 spikes**

**2. Cost curve (ref 502, solid line): Quadratic, starting near harvest level**

**- Formula:  $\text{Cost} = 0.225 + (0.006 \times s) + (0.0003 \times s^2)$  mWh/step**

**- Composed of base consumption ( $45 \text{ mW} \times 0.005 \text{ h} = 0.225 \text{ mWh}$ ) plus**

**linear activity cost ( $1.2 \text{ mW} \times 0.005 \text{ h} = 0.006 \text{ mWh/spike}$ ) plus**

**quadratic burst penalty ( $0.06 \times 0.005 = 0.0003 \text{ mWh/spike squared}$ )**

**- Rises from 0.225 mWh at 0 spikes to approximately 1.3 mWh at 50 spikes**

Crossover point (ref 506): Harvest equals Cost at approximately 0.3 spikes. Below this point (0 to 0.3 spikes), cost exceeds harvest (quiescent drain). Above this point (0.3 to 50+ spikes), harvest exceeds cost (net positive).

**Shaded region (ref 504, hatched): Area between curves where harvest exceeds**

**cost, labeled "NET POSITIVE REGION" with sublabel "Harvest >> Cost (Self-Sustaining)."**

Formula annotation box (ref 508): Contains exact formulas for both curves.

**Zone annotation at bottom: "Zone 1: Quiescent Drain — 0-1 Spikes: Cost > Harvest"**

Zero reference line: Horizontal dotted line at  $y = 0$ .

## **FIG. 6 — Hardware Reference Design**

This figure shows a top-down component layout diagram.

### **Central component:**

- **Raspberry Pi Pico (RP2040) 600, shown as a rectangle with pin labels**

**on edges**

### **Connected components (arranged around Raspberry Pi Pico 600):**

#### **Top left:**

- **Piezo Disc 27mm 602, shown as a circle**
- **Connected to Raspberry Pi Pico 600 via ADC pin (GP26/ADC0)**
- **Mechanical coupling line to SG90 Micro Servo 604**

#### **Top right:**

- **SG90 Micro Servo 604, shown as a rectangle**
- **Connected to Raspberry Pi Pico 600 via PWM pin (GP0)**

- Mechanical coupling to Piezo Disc 602 (zigzag line)

**Bottom left:**

- NTC 10K Thermistor 606, shown as a small component
- Connected to Raspberry Pi Pico 600 via ADC pin (GP27/ADC1)

**Bottom center:**

- GL5528 Photoresistor (LDR) 610, shown as a small component
- Connected to Raspberry Pi Pico 600 via ADC pin (GP28/ADC2)

**Bottom right:**

- WS2812B RGB LED 608, shown as a small square
- Connected to Raspberry Pi Pico 600 via digital pin (GP15)

**Left side:**

- Laird Ferrite Inductor 612
- Connected in series between Piezo Disc 602 and rectifier bridge 614

**- Label: "LC resonance circuit"**

**Power section:**

**- Rectifier bridge 614**

**- Storage capacitor 616**

**- Arrow labeled "Harvested energy feedback"**

**BOM annotation box in corner:**

**- "Raspberry Pi Pico (RP2040)"**

**- "SG90 Micro Servo"**

**- "27mm Piezoelectric Disc"**

**- "NTC 10K Thermistor"**

**- "Photoresistor (LDR)"**

**- "WS2812B RGB LED"**

**- "Misc (wires, board, inductor)"**



## FIG. 7 — Validation Results Summary

This figure comprises a validation results summary table 700 and an STDP learning efficacy bar chart 702 summarizing all validated claims.

Validation results summary table 700 with columns: Claim | Criterion |

Measured Value | Threshold | Result

Section header: "Phase 7 Control Baseline (EnergyConfig)"

Row A: Closed-loop continuity | All channels 2000 points | 2000 | 2000 | PASS

Row B: Boundedness | All state variables bounded | bounded | bounded | PASS

Row C: Robustness under noise | Std(theta) <= 45 deg | [value] | 45 | PASS

Row D: Input-output gain | corr(L,theta) >= 0.2 | [value] | 0.2 | PASS

Row E: Saturation ratio | sat\_theta <= 0.20 | [value] | 0.20 | PASS

Section header: "Phase 7 Full Integration (Control confirms energy depletion)"

Row: Energy at 2000 steps | Energy depleted | -4,697 mWh | < 0 | CONFIRMED

**Section header: "MCP Integration (BalancedEnergyConfig)"**

**Row: Energy at 2000 steps | Homeostasis at capacity | 100.0 mWh (capped) | > 0 |  
CONFIRMED**

**Section header: "Phase 8 STDP"**

**Row F8.1: Weight entropy decrease | final < initial | 2.95 to 2.02 bits | decrease | PASS**

**Row F8.2: STDP MI > 1.2x frozen | ratio | 6.52x | > 1.20 | PASS**

**Row F8.3: Late weight delta < 10% early | ratio | 0.00% | < 10% | PASS**

**STDP learning efficacy bar chart 702 presents the Phase 8 STDP results**

**graphically for visual comparison of STDP-enabled versus frozen-weight  
conditions.**

**Footer: "All claims validated. System is reproducible via git tags."**

**FIG. 8 — Energy-Bounded Recursive Control Architecture**

**This figure is the canonical embodiment and reference signal flow for the  
entire system. It shows the complete architecture as a vertically stacked**

**block diagram with bidirectional control paths and energy feedback. This is the primary reference figure for patent examination.**

**System operation is constrained by real-time energy availability derived from physical interaction with the environment, thereby ensuring bounded, non-abstract execution.**

**Layout — Vertically stacked functional blocks with lateral connections:**

**TOP — Physical Environment:**

The physical environment sits at the top of the diagram.

**Two downward arrows from the physical environment:**

**Left arrow: "(Light / Temp / Force)" — to Sensor Interface 802**

**Right arrow: "(Mechanical Load)" — to Energy Harvesting 804**

**TIER 1 — Dual Input (left and right, same level):**

**Left block:**

**Sensor Interface 802 receives environmental signals (light, temperature,**

**force) through specific sensor ICs: BH1750 for light, DS18B20 for**

temperature, and piezo input for force. Arrow down from Sensor Interface 802 to Neuromorphic Core 806.

**Right block:**

**Energy Harvesting 804 performs physical energy transduction: piezoelectric**

element through full-bridge rectifier to supercapacitor storage. Arrow down from Energy Harvesting 804 to Neuromorphic Core 806.

**TIER 2 — Central Processing (full width):**

**Neuromorphic Core (SNN) 806 occupies the full width of the diagram. It**

**contains four labeled subsystems:**

- "Spiking Neurons" — leaky integrate-and-fire neural population

- "Membrane Dynamics" —  $V(t) = V(t-1) \times \text{leak} + I_{\text{adjusted}}$

- "Optional STDP" — spike-timing dependent plasticity (opt-in)

- "Activity-Dependent Energy Cost" — quadratic:  $\text{base} + k \times (\text{spikes}/N)^2$

**Two downward arrows from Neuromorphic Core 806:**

**Left arrow: to Recursive Reflection 808**

**Right arrow: to Energy Governor 810**

**TIER 3 — Dual Processing (left and right, same level, BIDIRECTIONAL):**

**Left block:**

**Recursive Reflection 808 implements the self-observation mechanism. It**

**feeds aggregate output at timestep  $t$  back as input at timestep  $t+1$ , scaled**

**by the reflection coefficient. Recursive Reflection 808 maintains prior state memory, output feedback, and temporal context.**

**Right block:**

**Energy Governor 810 enforces energy constraints. It monitors storage**

**level, applies the activity-dependent cost function, and damps activity**

**when energy is low. Energy Governor 810 manages storage level, cost function, and activity damping.**

**CRITICAL: Bidirectional arrow between Recursive Reflection 808 and**

Energy Governor 810.

**Label: "Energy state modulates reflection gain; reflection state**

**influences energy cost through activity level"**

**Both Recursive Reflection 808 and Energy Governor 810 have arrows downward to Cognitive Modulation Layer 812.**

**TIER 4 — Modulation (full width):**

**Cognitive Modulation Layer 812 occupies the full width of the diagram. It**

**contains four labeled functions:**

- **"Attention Weighting" — prioritizes input channels**
- **"Intention Bias" — goal-directed modulation**
- **"Contextual Suppression / Amplification" — state-dependent gating**
- **"(Optional) External Advisory Input" — external control layer when connected**

**Two downward arrows from Cognitive Modulation Layer 812:**

**Left arrow: to Neural Router 814**

**Right arrow: to Telemetry & Logging 816**

**TIER 5 — Output (left and right, same level):**

**Left block:**

**Neural Router 814 performs signal arbitration. It routes neural output to**

**actuators based on priority, gates signals when energy is insufficient,**

and prevents output saturation through congestion control. Arrow down from Neural Router 814 to Actuation & Response 818.

**Right block:**

**Telemetry & Logging 816 provides observability output. It records energy**

level, system state, and reflection metrics for validation and monitoring. No downward arrow from Telemetry & Logging 816 (terminal output).

**TIER 6 — Actuation (full width):**

**Actuation & Response 818 occupies the full width of the diagram. It**

**contains servo control, mechanical interaction, and environmental**

**perturbation subsystems. Actuation & Response 818 drives mechanical**

**interaction, which perturbs the environment (closing the loop back to**

**the physical environment at top) and generates mechanical load (feeding**

**Energy Harvesting 804 at Tier 1).**

**Two upward feedback arrows from Actuation & Response 818:**

**1. Arrow back to the physical environment (top): "Environmental Perturbation"**

**2. Arrow back to Energy Harvesting 804 (Tier 1 right): "Mechanical Load**

**generates piezoelectric energy"**

**FEEDBACK LOOPS (mark each distinctly):**

**Loop A — Energy Harvesting Loop:**

**Actuation & Response 818 → Mechanical Load → Energy Harvesting 804 →**

**Neuromorphic Core 806**

**Style: Bold solid arrows**



**Label: "ENERGY LOOP — Motor activity generates power for neural computation"**

**Loop B — Recursive Reflection Loop:**

**Neuromorphic Core 806 → Recursive Reflection 808 → Neuromorphic Core 806**

**Style: Dotted arrows**

**Label: "REFLECTION LOOP — Prior state observation at configurable gain"**

**Loop C — Environmental Perturbation Loop:**

**Actuation & Response 818 → Environment → Sensor Interface 802 →**

**Neuromorphic Core 806**

**Style: Dashed arrows**

**Label: "SENSORY LOOP — Actions change environment, environment changes input"**

**Loop D — Energy Governor Feedback:**

**Energy Harvesting 804 → Energy Governor 810 ↔ Recursive Reflection 808 →**

## **Cognitive Modulation Layer 812**

**Style: Thin solid arrows**

**Label: "REGULATION LOOP — Energy availability constrains all processing"**

**Key annotations at bottom of figure:**

**"This architecture is physically grounded: all computation is constrained**

**by real-time energy availability derived from the system's own mechanical**

**interaction with its environment."**

**"No external power source is required for sustained operation under the**

**balanced energy configuration."**

**FIG. 8 illustrates one exemplary embodiment; alternative embodiments may**

**omit, combine, or substitute individual modules while preserving the**

**energy-bounded recursive control loop.**

**Sheet numbering: "8/8" at top center**

## **GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)**

- **Paper size: 8.5 x 11 inches (US Letter)**
- **Top margin: 1 inch**
- **Left margin: 1 inch**
- **Right margin: 5/8 inch**
- **Bottom margin: 3/8 inch**
- **Black lines on white background ONLY**
- **No color, no grayscale shading (use hatching patterns instead)**
- **Line weight: sufficiently heavy for adequate reproduction**
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/8", "2/8", etc.
- **No frames or borders around drawing area**
- **All text in figures must be at least 1/8 inch (3.2 mm) high**
- **Reference numerals must be used consistently across all figures**